

Error in shared pool

ORA-04031: unable to allocate 16272 bytes of shared memory ("shared pool","unknown object","sga heap(2,0)","ksu:stats_freelist")

ORA-04031: Oracle tried to allocate memory in the Shared Pool but failed.

16272 bytes: the amount of memory it tried to allocate.

"shared pool": the memory area where the allocation failed.

"unknown object": the exact object type Oracle was trying to allocate is unknown.

"sga heap(2,0)": the specific heap within the SGA where the problem occurred.

"ksu:stats_freelist": the internal structure or object type related to statistics or temporary objects that needed memory.

In short: Oracle needed ~16 KB more memory in the Shared Pool but there wasn't enough free space → this caused ORA-04031.

Automatic Shared Memory Management (ASMM):

- ASMM is active, so Oracle dynamically manages the size of the Shared Pool within the SGA limits.
- Despite ASMM, this error can occur if a request for memory exceeds the available free memory before Oracle can expand the pool.

--check memory

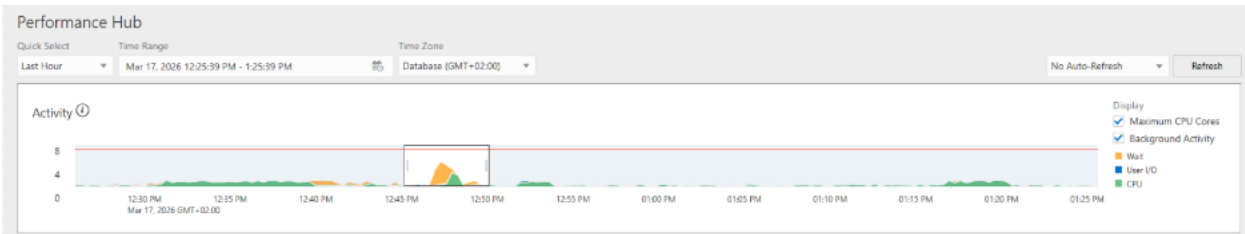
	total	used	free	shared	buff/cache	available
Mem:	47Gi	20Gi	1.7Gi	22Mi	25Gi	26Gi
Swap:	15Gi	925Mi	15Gi			

--the safest Memory size is 48GB

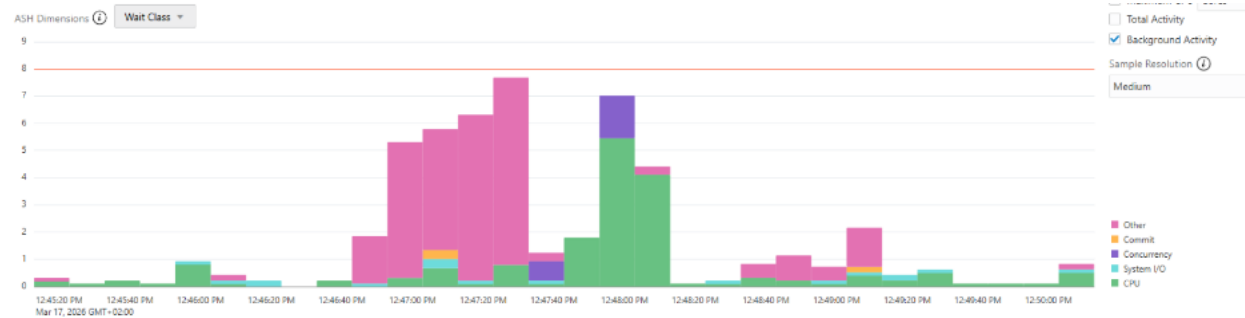
From OEM

select DB -> performance hup -> ASH Analytics

-- Activity



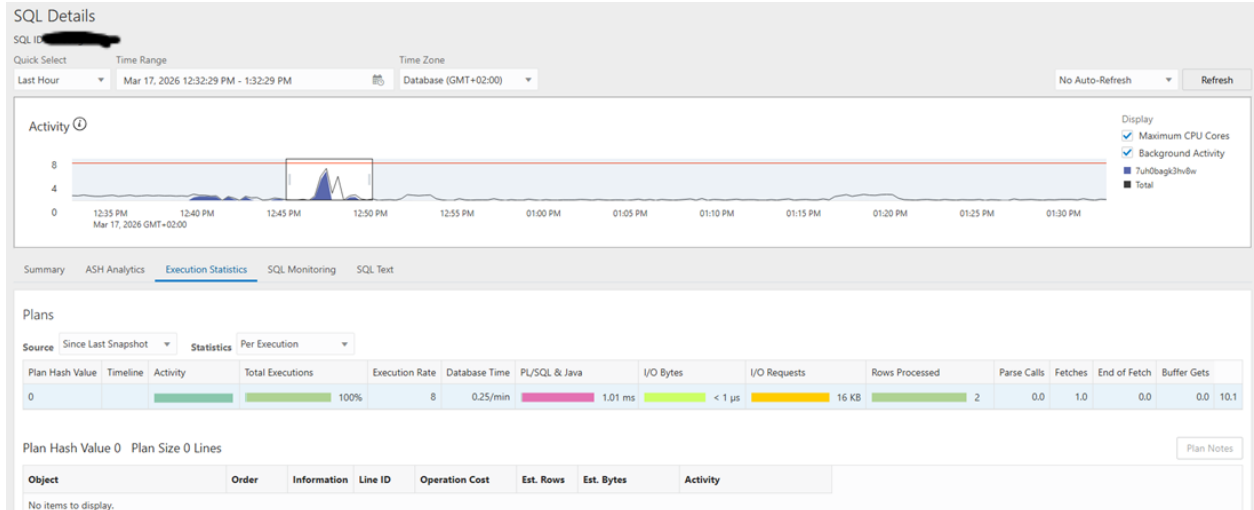
-- Average Active Sessions



Wait Class -> SQL Details

SQL ID	Activity (Average Active Sessions)	SQL Plan Hash Value	SQL Type	Service
[REDACTED]	0.89	N/A	PL/SQL EXECUTE	[REDACTED]
SQL ID: [REDACTED], SQL Text: BEGIN dbms_session.set_role(:1); END;				
[REDACTED]	0.08	27270497	SELECT	[REDACTED]
[REDACTED]	0.03	N/A	PL/SQL EXECUTE	[REDACTED]
[REDACTED]	0.03	3752461848	SELECT	[REDACTED]
[REDACTED]	0.02	N/A	PL/SQL EXECUTE	[REDACTED]
[REDACTED]	< 0.01	1911072918	SELECT	SYSS\$BACKGROUND

BEGIN dbms_session.set_role(:1); END;



Wait Class	by Wait Class	User Session	by Wait Class
Wait Class	Activity (Average Active Sessions)	User Session	Activity (Average Active Sessions) User Name Program Service
Other	0.89	1:11330.61993	< 0.01 [REDACTED] JDBC Thin Client [REDACTED]
CPU	< 0.01	1:5698.54184	< 0.01 [REDACTED] JDBC Thin Client [REDACTED]
		1:3818.46172	< 0.01 [REDACTED] JDBC Thin Client [REDACTED]
		1:1942.43518	< 0.01 [REDACTED] JDBC Thin Client [REDACTED]
		1:65.62820	< 0.01 [REDACTED] JDBC Thin Client [REDACTED]
		1:13212.41209	< 0.01 [REDACTED] JDBC Thin Client [REDACTED]
		1:5710.37707	< 0.01 [REDACTED] JDBC Thin Client [REDACTED]
		1:7591.45115	< 0.01 [REDACTED] JDBC Thin Client [REDACTED]
		1:77.33931	< 0.01 [REDACTED] JDBC Thin Client [REDACTED]
		1:3829.48210	< 0.01 [REDACTED] JDBC Thin Client [REDACTED]
		Others	0.86

```

SQL> COLUMN pool FORMAT A15
COLUMN name FORMAT A30
COLUMN current_size_gb FORMAT 999.9999
COLUMN min_size_gb FORMAT 999.9999
COLUMN max_size_gb FORMAT 999.9999
COLUMN size_gb FORMAT 999.9999
SELECT component AS pool,
       ROUND(current_size/1024/1024/1024,4) AS current_size_gb,
       ROUND(min_size/1024/1024/1024,4) AS min_size_gb,
       ROUND(max_size/1024/1024/1024,4) AS max_size_gb
FROM v$sga_dynamic_components
WHERE component='shared pool'; SQL> SQL> SQL> SQL> SQL> SQL> 2

```

POOL	CURRENT_SIZE_GB	MIN_SIZE_GB	MAX_SIZE_GB
shared pool	2.3750	2.1875	2.3750

```
SQL> sho parameter sga
```

NAME	TYPE	VALUE
allow_group_access_to_sga	boolean	FALSE
lock_sga	boolean	FALSE
pre_page_sga	boolean	TRUE
sga_max_size	big integer	18G
sga_min_size	big integer	0
sga_target	big integer	17G
unified_audit_sga_queue_size	integer	1048576

```
SQL> sho parameter pga
```

NAME	TYPE	VALUE
pga_aggregate_limit	big integer	100G
pga_aggregate_target	big integer	4G

```
SQL> 
```

svr-dbb-03.must.edu.eg

4%

20.74 GB / 47.21 GB

5.09 Mb/s

```
SQL> SELECT name,
        ROUND(bytes/1024/1024,2) AS size_mb
FROM v$sgastat
WHERE pool='shared pool' AND name='free memory'; 2    3    4

NAME                                SIZE_MB
-----
free memory                        181

SQL> SELECT name,
        ROUND(bytes/1024/1024/1024,4) AS size_gb
FROM v$sgastat
WHERE pool='shared pool'
ORDER BY bytes DESC
FETCH FIRST 10 ROWS ONLY; 2    3    4    5    6

NAME                                SIZE_GB
-----
ksunfy_meta 1                      .3732
private strands                    .2098
free memory                        .1720
session                            .1429
db_block_hash_buckets              .1250
SQLA                                .1209
KTI-UNDO                           .1202
S0 private sga                     .1196
KGLH0                               .0625
KSRMA State Object                 .0594

10 rows selected.

SQL> 
```

--Shared Pool current size: ~2.375 GB

--Free memory: ~262 MB

=====queries=====

COLUMN pool FORMAT A15

COLUMN name FORMAT A30

COLUMN current_size_gb FORMAT 999.9999

COLUMN min_size_gb FORMAT 999.9999

COLUMN max_size_gb FORMAT 999.9999

COLUMN size_gb FORMAT 999.9999

-- [1] Shared Pool sizes

```
SELECT component AS pool, ROUND(current_size/1024/1024/1024,4) AS current_size_gb, ROUND(min_size/1024/1024/1024,4) AS  
min_size_gb, ROUND(max_size/1024/1024/1024,4) AS max_size_gb FROM v$sga_dynamic_components WHERE component='shared  
pool';
```

-- **2** Check largest consumers in Shared Pool

```
SELECT name, ROUND(bytes/1024/1024/1024,4) AS size_gb FROM v$sgastat WHERE pool='shared pool' ORDER BY bytes DESC FETCH  
FIRST 10 ROWS ONLY;
```

-- **3** free memory inside Shared Pool

```
SELECT name, ROUND(bytes/1024/1024,2) AS size_mb FROM v$sgastat WHERE pool='shared pool' AND name='free memory';
```